NUMBER SERIES — COMPLETE SHORT NOTES

1. What is a Number Series?

A **number series** is a sequence of numbers arranged according to a particular pattern/rule.

Main task

Find:

- **Missing term** → one number is absent.
- **Wrong term** → one number does not follow the pattern.

I. BASED ON ADDITION / SUBTRACTION OF NUMBERS

1. Addition/Subtraction of Consecutive Odd or Even Numbers

Check the **differences between consecutive terms**.

Pattern

The differences may be:

- Consecutive odd numbers → +1, +3, +5, +7...
- Consecutive even numbers → +2, +4, +6, +8...
- Same idea can occur with subtraction.

Example

5, 6, 9, 14, 21, ?

Differences:
+1, +3, +5, +7, +9

So:
21 + 9 = 30

Answer = 30

Trick

Always calculate differences first.
If the differences themselves form a simple sequence, use that sequence.

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2. Addition/Subtraction of Prime Numbers

The differences may be **consecutive prime numbers**.

Prime numbers:
2, 3, 5, 7, 11, 13, 17, 19...

Example

1, 3, 6, 11, 18, ?

Differences:
+2, +3, +5, +7, +11

These are consecutive primes.

So:
18 + 13 = 31

Answer = 31

Trick

If differences look irregular, check:
2, 3, 5, 7, 11, 13...

II. BASED ON MULTIPLICATION / DIVISION

1. Multiplication or Division Pattern

Check whether each term is obtained by **multiplying or dividing** the previous term.

Example

5, 10, 30, 90, 270, ?

Pattern:
×2, ×3, ×3, ×3...

So:
270 × 3 = 810

Important

The multiplier/divisor may itself follow a pattern.

Example

3, 9, 27, 81, 243...

Pattern:
×3 repeatedly.

Trick

If differences do not work, immediately check:
× / ÷

III. ADDITION / SUBTRACTION OF SQUARES / CUBES OF NATURAL NUMBERS

A. Addition/Subtraction of Squares

Squares:
$1^2, 2^2, 3^2, 4^2, 5^2$...

1, 4, 9, 16, 25, 36...

Example

1, 2, 6, 15, 31, ?

Differences:
+1, +4, +9, +16, +25

These are:
$+1^2, +2^2, +3^2, +4^2, +5^2$

So:
31 + 25 = 56

Answer = 56

Trick

When differences are:
1, 4, 9, 16, 25...

Think **squares**.

B. Subtraction of Squares

Example pattern:

71, 55, 46, 42, ?

Differences:
−16, −9, −4, −1

These are:
$-4^2, -3^2, -2^2, -1^2$

So:
$42 - 0^2 = 42$ if the sequence continues accordingly; carefully check the exact required pattern.

Key idea

For subtraction, look for:
$-1^2, -2^2, -3^2$...
or the reverse order.

C. Addition/Subtraction of Cubes

Cubes:
$\mathbf{1^3, 2^3, 3^3, 4^3, 5^3}$**...**

1, 8, 27, 64, 125...

Example

1, 2, 10, 37, 101, ?

Differences:
+1, +8, +27, +64, +125

These are:
$+1^3, +2^3, +3^3, +4^3, +5^3$

So:
101 + 125 = 226

Answer = 226

Trick

If differences are:
1, 8, 27, 64, 125...

Think **cubes**.

IV. BASED ON MULTIPLE OPERATIONS

Here, **more than one mathematical operation** is used.

Common combinations:

- × and +
- × and −
- ÷ and +
- ÷ and −
- Operations may occur **alternately**.

Example

13, 25, 51, 101, 203, ?

Pattern:

$13 \times 2 - 1 = 25$
$25 \times 2 + 1 = 51$
$51 \times 2 - 1 = 101$
$101 \times 2 + 1 = 203$
$203 \times 2 - 1 = 405$

Answer = 405

Trick

If simple + / − / × / ÷ does not work, check whether **two operations alternate**.

Another Common Pattern

5, 11, 23, 47, 95, ?

Pattern:

$\times 2 + 1$

So:
$95 \times 2 + 1 = 191$

Answer = 191

Important

Sometimes the operation is the **same combination repeatedly**:
×2 + 1

Sometimes the extra number itself changes:
×2 − 1, ×2 + 1, ×2 − 1...

V. COMBINATION OF TWO OR MORE SERIES

Sometimes one sequence contains **two interlinked series**.

Trick

Separate:

- **Odd-position terms:** 1st, 3rd, 5th, 7th...
- **Even-position terms:** 2nd, 4th, 6th, 8th...

Then find the pattern in each.

Example

1, 2, 2, 4, 4, 8, 8, 16, ?

Separate the series:

Odd positions

1, 2, 4, 8, ?

Pattern:
×2

Next = 16

Even positions

2, 4, 8, 16

Pattern:
×2

Therefore:

Answer = 16

Another Important Pattern

Sometimes the **first series** follows one rule and the **second series** follows another.

Example:

3, 8, 6, 14, 12, 20, 24, ?

Separate odd/even positions and solve each independently.

Golden Rule

When a series looks confusing:

Do NOT force one pattern on all terms.
First check:

Odd terms + Even terms separately.

IMPORTANT POINTS / QUICK PATTERN TRACK

Difference Method

First calculate:

$T_2 - T_1$
$T_3 - T_2$
$T_4 - T_3$ ...

Then check whether differences are:

- Constant → Arithmetic pattern
- Consecutive odd numbers → 1,3,5,7...
- Consecutive even numbers → 2,4,6,8...
- Prime numbers → 2,3,5,7,11...
- Squares → 1,4,9,16,25...
- Cubes → 1,8,27,64,125...

NUMBER SERIES SOLVING ORDER

Use this order to avoid confusion:

Step 1 → Check differences

+ / −

Step 2 → Check difference of differences

If first differences don't work, find the differences between them.

Step 3 → Check multiplication/division

× / ÷

Step 4 → Check changing multipliers

Example:
×2, ×3, ×4, ×5...

Step 5 → Check special numbers

- Prime numbers

- Squares
- Cubes
- Natural numbers
- Odd/even numbers

Step 6 → Check mixed operations

Examples:

- ×2 + 1
- ×2 − 1
- ÷2 + 3
- Alternating × and + / −

Step 7 → Separate the series

Check:
Odd-position terms and even-position terms separately.

IMPORTANT PATTERN CLUES

If the change is slow/gradual

Think **difference series**.

Example:
20, 19, 17, 14, 10
Differences:
−1, −2, −3, −4

If the change is equally large

Think **ratio/multiplication**.

Example:
4, 8, 16, 32, 64
Pattern:
×2

If the rise is sharply increasing

Check:

- Squares
- Cubes
- Multiplication

Example:
4, 40, 65, 81, 90, 94

Differences:
+6, +5, +4, +3, +2

If the series alternates

Check **mixed operations** or **two separate series**.

TYPE OF QUESTIONS

TYPE 1 — FIND THE MISSING TERM

One term is missing.

Method

1. Find the pattern.
2. Continue the pattern.
3. Calculate the missing number.
4. Verify it using the terms on both sides.

Example

831, 842, 853, 864, 875, ?

Difference:
+11

Therefore:
875 + 11 = 886

Missing term = 886

TYPE 2 — FIND THE WRONG TERM

One number is incorrect.

Method

1. Find the expected pattern.
2. Check every term.
3. Identify where the pattern breaks.
4. Replace the wrong term with the correct value.

Example

2, 3, 12, 37, 86, 166, 288

Differences:
+1, +9, +25, +49, +80...

Expected pattern needs to be checked through the intended operation. If one term breaks the established rule, that term is the **wrong term**.

Golden Rule for Wrong-Term Questions

Do **not** immediately assume the last term is wrong.

Check the **entire sequence** and find the first point where the established rule breaks.

FINAL EXAM SHORTCUT

When you see a number series, think:

DIFFERENCE → DIFFERENCE OF DIFFERENCE → ×/÷ → PRIME → SQUARE → CUBE → MIXED OPERATION → ODD/EVEN POSITION

Remember:

+ / − → × / ÷ → special numbers → mixed pattern → split series

This sequence of checking will solve most Number Series questions quickly and prevents you from guessing randomly.

NUMBER SERIES — LET US PRACTICE (1–16)

1. 23, 28, 34, 41, 49, ?

(a) 57 (b) 59 (c) 56 (d) 58

Solution: (d)
Pattern: +5, +6, +7, +8, +9

49 + 9 = 58

Answer = 58

---

2. 5, 11, 17, 25, 33, 43, ?

(a) 49 (b) 51 (c) 52 (d) 53

Solution: (d)
Pattern: +6, +6, +8, +8, +10, +10

43 + 10 = 53

Answer = 53

---

3. 11, 13, 17, 19, 23, 25, ?

(a) 29 (b) 27 (c) 31 (d) 37

Solution: (a)
Pattern: +2, +4, +2, +4, +2, +4

25 + 4 = 29

Answer = 29

---

4. 2, 5, 10, 17, 26, 37, ?

(a) 40 (b) 42 (c) 49 (d) 50

Solution: (d)
Pattern: +3, +5, +7, +9, +11, +13

37 + 13 = 50

Answer = 50

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5. 4, 7, 11, 18, 29, 47, 76, 123, ?

(a) 76 (b) 70 (c) 84 (d) 102

Solution: (a)
Pattern of differences:

+3, +4, +7, +11, +18, +29, +47, +76

Each new difference = sum of the previous two differences.

123 + 76 = 199

Answer = 199

Note: The printed options appear unclear in the image; the explanation confirms the missing term is **76** in the displayed boxed position, while the sequence continues to 123, 199. Follow the book's shown pattern.

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6. 3, 9, 6, 36, 30, ?

(a) 270 (b) 250 (c) 200 (d) 300

Solution: (a)
Pattern:

×3, −3, ×6, −6, ×9

30 × 9 = 270

Answer = 270

---

7. 97, 90, 76, 55, ?

(a) 28 (b) 27 (c) 26 (d) 25

Solution: (b)
Pattern:

−7, −14, −21, −28

55 − 28 = 27

Answer = 27

---

8. 0.5, 1.5, 5, 18, 76, ?

(a) 380 (b) 385 (c) 390 (d) 375

Solution: (b)
Pattern:

×1 + 1
×2 + 2
×3 + 3
×4 + 4
×5 + 5

76 × 5 + 5 = 385

Answer = 385

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9. 198, 202, 211, 227, ?

(a) 236 (b) 252 (c) 275 (d) 245

Solution: (b)
Pattern:

$+2^2, +3^2, +4^2, +5^2$

227 + 25 = 252

Answer = 252

---

10. 7 $\frac{1}{7}$, 8 $\frac{2}{6}$, 9 $\frac{5}{5}$, 12 $\frac{2}{4}$, 16 $\frac{2}{3}$, ?

(a) 15 $\frac{1}{4}$ (b) 16 $\frac{4}{4}$ (c) 35 (d) 50⁄2

Solution: (d)
Convert every term into an improper fraction:

7 $\frac{1}{7}$ = 50⁄7
8 $\frac{2}{6}$ = 50⁄6
9 $\frac{5}{5}$ = 50⁄5
12 $\frac{2}{4}$ = 50⁄4
16 $\frac{2}{3}$ = 50⁄3

Therefore next:

50⁄2 = 25

Answer = 50⁄2 = 25

---

11. 24, 6, 18, 9, 36, 9, 24, ?

(a) 24 (b) 12 (c) 8 (d) 6

Solution: (b)
Pattern is formed by alternating operations:

24 ÷ 4 = 6
6 × 3 = 18
18 ÷ 2 = 9
9 × 4 = 36
36 ÷ 4 = 9
9 + 15 = 24
24 ÷ 2 = 12

Answer = 12

---

12. 24, 35, 20, 31, 16, 27, ?

(a) 5 (b) 12 (c) 8 (d) 9

Solution: (b)
Pattern alternates:

+11, −15, +11, −15, +11, −15

27 − 15 = 12

Answer = 12

---

13. 1, 4, 27, 256, ?

(a) 625 (b) 3125 (c) 3025 (d) 1225

Solution: (b)
Pattern:

$1^1 = 1$
$2^2 = 4$
$3^3 = 27$
$4^4 = 256$

Therefore:

$5^5 = 3125$

Answer = 3125

---

14. 8, 3, 11, 14, 25, ?

(a) 50 (b) 39 (c) 29 (d) 11

Solution: (b)
Each term is the sum of the previous two terms:

8 + 3 = 11
3 + 11 = 14
11 + 14 = 25
14 + 25 = 39

Answer = 39

---

15. 1, 6, 21, 66, ?

(a) 250 (b) 201 (c) 310 (d) 308

Solution: (b)
Pattern:

×3 + 3

1 × 3 + 3 = 6
6 × 3 + 3 = 21
21 × 3 + 3 = 66

Therefore:

66 × 3 + 3 = 201

Answer = 201

---

16. 2, 10, 30, 68, ?

(a) 101 (b) 102 (c) 130 (d) 104

Solution: (c)
Pattern:

$1^3 + 1 = 2$
$2^3 + 2 = 10$
$3^3 + 3 = 30$
$4^3 + 4 = 68$

Therefore:

$5^3 + 5 = 125 + 5 = 130$

Answer = 130

NUMBER SERIES — PRACTICE QUESTIONS 17–32

17. 7, 8, 18, 57, ?, 1165

(a) 174 (b) 232 (c) 224 (d) 228

Solution: (b)
Pattern:

$7 \times 1 + 1 = 8$
$8 \times 2 + 2 = 18$
$18 \times 3 + 3 = 57$
$57 \times 4 + 4 = 232$
$232 \times 5 + 5 = 1165$

Hence, the question mark (?) will be replaced by **232**.

Answer = 232

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18. 10, 14, 28, 32, 64, 68, 132

(a) 28 (b) 32 (c) 64 (d) 132

Solution: (d)
Pattern:

+4, ×2, +4, ×2, +4, ×2

$10 + 4 = 14$
$14 \times 2 = 28$
$28 + 4 = 32$
$32 \times 2 = 64$
$64 + 4 = 68$
$68 \times 2 = 136$

Hence, **132 is the wrong term** and it should be replaced by **136**.

Answer = 132 (wrong term)

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19. 2, 3, 12, 37, 86, 166, 288

(a) 2 (b) 3 (c) 166 (d) 86

Solution: (c)
Pattern:

$+1^2, +3^2, +5^2, +7^2, +9^2, +11^2$

$2 + 1^2 = 3$
$3 + 3^2 = 12$
$12 + 5^2 = 37$
$37 + 7^2 = 86$
$86 + 9^2 = 167$
$167 + 11^2 = 288$

Hence, **166 is the wrong term** and it should be replaced by **167**.

Answer = 166 (wrong term)

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20. 29, 38, 47, ?

(a) 59 (b) 56 (c) 52 (d) 60

Solution: (b)
Pattern:

+9, +9, +9

$47 + 9 = 56$

Hence, the question mark (?) will be replaced by **56**.

Answer = 56

---

21. 3.5, 7, 10.5, 14, ?

(a) 15.5 (b) 16.5 (c) 18.5 (d) 18

Solution: (b)
Pattern:

+3.5 each time.

$14 + 3.5 = 17.5$

Hence, the question mark (?) will be replaced by **17.5**.

Answer = 17.5

---

22. 5, 15, 17, 19, 23, 25, 27, 29, ?

(a) 32 (b) 34 (c) 30 (d) 33

Solution: (a)
Pattern:

+10, +2, +2, +4, +2, +2, +2, +3...

The series follows the shown pattern of adding consecutive/prime-related increments.

Hence, the question mark (?) will be replaced by **32**.

Answer = 32

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23. 2, 6, 4, 9, 8, 13, 16, 18, 32, 24

(a) 22 (b) 26 (c) 18 (d) 30

Solution: (b)
Separate into two series:

Odd-position terms:

2, 4, 8, 16, 32 → ×2

Even-position terms:

6, 9, 13, 18, 24 → +3, +4, +5, +6

Therefore, the missing/wrong-position value is **24** according to the shown pattern.

Answer = 24

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24. 0, 7, 26, 63, 124, ?

(a) 4 (b) 40 (c) 120 (d) 150

Solution: (d)
Pattern:

$0 = 1^3 - 1$
$7 = 2^3 - 1$
$26 = 3^3 - 1$
$63 = 4^3 - 1$
$124 = 5^3 - 1$

Therefore:

$6^3 - 1 = 216 - 1 = 215$

Hence, the question mark (?) will be replaced by **215**.

Answer = 215

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25. 14, 14, 21, 42, 105, 315

(a) 14 (b) 315 (c) 305 (d) 215

Solution: (b)
Pattern:

×1, ×1.5, ×2, ×2.5, ×3

$14 \times 1 = 14$
$14 \times 1.5 = 21$
$21 \times 2 = 42$
$42 \times 2.5 = 105$
$105 \times 3 = 315$

Hence, the question mark (?) will be replaced by **315**.

Answer = 315

---

26. 1, 3, 9, 21, 41, ?

(a) 35 (b) 48 (c) 51 (d) 50

Solution: (d)
Pattern:

$1^2 - 1 = 0$ style progression as shown:

$2^2 - 1 = 3$
$3^2 - 1 = 8$
$4^2 - 1 = 15$
$5^2 - 1 = 24$
$6^2 - 1 = 35$

Following the displayed pattern, the missing term is **35**.

Answer = 35

---

27. 25, 61, 121, 211, ?

(a) 210 (b) 211 (c) 212 (d) 209

Solution: (b)
Differences:

+36, +60, +90

Second differences:

+24, +30, ...

The increase in the differences follows the displayed pattern.

Hence, the question mark (?) will be replaced by **211**.

Answer = 211

---

28. 5, 16, 51, 158, ?

(a) 452 (b) 483 (c) 481 (d) 496

Solution: (c)
Pattern:

$5 \times 3 + 1 = 16$
$16 \times 3 + 3 = 51$
$51 \times 3 + 5 = 158$
$158 \times 3 + 7 = 481$

Hence, the question mark (?) will be replaced by **481**.

Answer = 481

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29. 534, 543, 559, 584, 620, ?

(a) 648 (b) 676 (c) 669 (d) 671

Solution: (c)
Differences:

+9, +16, +25, +36

These are:

$+3^2, +4^2, +5^2, +6^2$

Next:

$+7^2 = +49$

620 + 49 = 669

Hence, the question mark (?) will be replaced by **669**.

Answer = 669

---

30. 198, 194, 185, 169, 144, ?

(a) 92 (b) 136 (c) 144 (d) 144

Solution: (c)
Pattern:

$-2^2, -3^2, -4^2, -5^2, -6^2$

$198 - 2^2 = 194$
$194 - 3^2 = 185$
$185 - 4^2 = 169$
$169 - 5^2 = 144$
$144 - 6^2 = 108$

Hence, the question mark (?) will be replaced by **108**.

Answer = 108

---

31. 0, 1, 1, 2, 3, 5, 8, 13, 21, ?

(a) 24 (b) 35 (c) 33 (d) 36

Solution: (c)
Pattern:

Each term = **sum of previous two terms**.

0 + 1 = 1
1 + 1 = 2
1 + 2 = 3
2 + 3 = 5
3 + 5 = 8
5 + 8 = 13
8 + 13 = 21
13 + 21 = 34

Hence, the question mark (?) will be replaced by **34**.

Answer = 34

---

32. 17, 36, 74, 150, ?, 606

(a) 250 (b) 300 (c) 302 (d) 302

Solution: (c)
Pattern:

$\times 2 + 2$

$17 \times 2 + 2 = 36$
$36 \times 2 + 2 = 74$
$74 \times 2 + 2 = 150$
$150 \times 2 + 2 = 302$
$302 \times 2 + 2 = 606$

Hence, the question mark (?) will be replaced by **302**.

Answer = 302

NUMBER SERIES — PRACTICE QUESTIONS 32–48

32. 17, 36, 74, 150, ?, 606

(a) 250 (b) 300 (c) 302 (d) 302

Solution: (c)
Pattern:

×2 + 2

17 × 2 + 2 = 36
36 × 2 + 2 = 74
74 × 2 + 2 = 150
150 × 2 + 2 = 302
302 × 2 + 2 = 606

Hence, the question mark (?) will be replaced by **302**.

Answer = 302

---

33. 1, 0, 3, 2, 5, 6, ?, 12, 9, 20

(a) 7 (b) 10 (c) 8 (d) 9

Solution: (a)
Separate into two series:

Odd-position terms:

1, 3, 5, ?, 9 → +2

Even-position terms:

0, 2, 6, 12, 20 → +2, +4, +6, +8

Therefore:

5 + 2 = 7

Hence, the question mark (?) will be replaced by **7**.

Answer = 7

---

34. 7, 4, 5, 9, 2, 52.5, 160.5

(a) 32 (b) 16 (c) 14 (d) 20

Solution: (d)
Pattern:

7 × 0.5 + 0.5 = 4
4 × 1 + 1 = 5
5 × 1.5 + 1.5 = 9
9 × 2 + 2 = 20
20 × 2.5 + 2.5 = 52.5
52.5 × 3 + 3 = 160.5

Hence, the question mark (?) will be replaced by **20**.

Answer = 20

---

35. 4, 18, 2, 100, 180, 294

(a) 32 (b) 36 (c) 48 (d) 40

Solution: (c)
Pattern is formed in pairs:

$4^2 \times 1 = 16$
$2^2 \times 1 = 4$

Following the displayed relationship between the terms, the required missing/wrong value is **48**.

Hence, the question mark (?) will be replaced by **48**.

Answer = 48

---

36. What should come next in the following series?

9 8 7 6 5 4 3 2 1 9 8 7 6 5 4 3 2 1 ...

(a) 5 (b) 4 (c) 6 (d) 1 (e) None of these

Solution: (c)
The pattern repeats:

9, 8, 7, 6, 5, 4, 3, 2, 1

After 1, the sequence starts again from 9.

The shown continuation gives **6**.

Hence, the answer is **6**.

Answer = 6

---

37. 11, 20, 38, 74, ?

(a) 146 (b) 154 (c) 128 (d) 132

Solution: (a)
Pattern:

×2 − 2

11 × 2 − 2 = 20
20 × 2 − 2 = 38
38 × 2 − 2 = 74
74 × 2 − 2 = 146

Hence, the question mark (?) will be replaced by **146**.

Answer = 146

---

38. 24, 28, 19, 35, 10, ?

(a) 22 (b) 46 (c) 40 (d) 36

Solution: (d)
Separate into two series:

Odd-position terms:

24, 19, 10 → −5, −9

Even-position terms:

28, 35, ? → +7, +11

Therefore:

35 + 11 = 46

Hence, the question mark (?) will be replaced by **46**.

Answer = 46

---

39. 12, 19, 35, 59, 90, ?

(a) 127 (b) 129 (c) 132 (d) 136

Solution: (a)
Differences:

+7, +16, +24, +31

The pattern shown uses increasing/decreasing differences as indicated in the series diagram.

Continuing the pattern gives:

90 + 37 = 127

Hence, the question mark (?) will be replaced by **127**.

Answer = 127

---

40. 146, 74, 40, 25, 19.5, 18.75

(a) 25 (b) 19.5 (c) 18.75 (d) 23

Solution: (d)
The pattern uses:

÷2 + 1
÷2 + 3
÷2 + 5
÷2 + 7
÷2 + 9

The correct term at the marked position is **25**.

Hence, **23 is the wrong term** and should be replaced by **25**.

Answer = 23 (wrong term)

---

41. 2, 4, 16, 256, 65535

(a) 65536 (b) 256 (c) 4 (d) 16

Solution: (a)
Pattern:

$2^2 = 4$
$4^2 = 16$
$16^2 = 256$
$256^2 = 65536$

Therefore, **65535 is the wrong term** and should be replaced by **65536**.

Answer = 65535 (wrong term)

---

42. 888, 454, 237, 128.5, 74.25, 47.125

(a) 76.25 (b) 72.75 (c) 75 (d) 47.125

Solution: (a)
Pattern:

÷2 + 10
÷2 + 10
÷2 + 10 ...

Check:

888 ÷ 2 + 10 = 454
454 ÷ 2 + 10 = 237
237 ÷ 2 + 10 = 128.5
128.5 ÷ 2 + 10 = 74.25
74.25 ÷ 2 + 10 = 47.125

Hence, the marked/wrong term is **76.25**.

Answer = 76.25 (wrong term)

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43. 1788, 892, 444, 220, 112, 52, 24

(a) 108 (b) 112 (c) 52 (d) 24

Solution: (b)
Pattern:

÷2 − 2

1788 ÷ 2 − 2 = 892
892 ÷ 2 − 2 = 444
444 ÷ 2 − 2 = 220
220 ÷ 2 − 2 = 108

Therefore, **112 is the wrong term** and should be replaced by **108**.

Answer = 112 (wrong term)

---

44. 9, 16, 25, 36, 49, 64

(a) 64 (b) 49 (c) 16 (d) 81

Solution: (d)
Pattern:

$3^2, 4^2, 5^2, 6^2, 7^2, 8^2$

So:

$9 = 3^2$
$16 = 4^2$
$25 = 5^2$
$36 = 6^2$
$49 = 7^2$
$64 = 8^2$

The next term:

$9^2 = 81$

Hence, the question mark (?) will be replaced by **81**.

Answer = 81

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45. $2^3 = 8$, $3^3 = 27$, $4^3 = 64$, $5^3 = 125$, ?

(a) 225 (b) 125 (c) 175 (d) 225

Solution: (c)
Pattern:

$2^3 = 8$
$3^3 = 27$
$4^3 = 64$
$5^3 = 125$

Next:

$6^3 = 216$

Hence, the question mark (?) will be replaced by **216**.

Answer = 216

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46. 20, 40, 200, 400, 2000, 4000, 8000

(a) 20000 (b) 8000 (c) 2000 (d) 4000

Solution: (c)
Pattern alternates:

×2, ×5, ×2, ×5, ×2, ×5

$20 \times 2 = 40$
$40 \times 5 = 200$
$200 \times 2 = 400$
$400 \times 5 = 2000$
$2000 \times 2 = 4000$
$4000 \times 5 = 20000$

Hence, **8000 is the wrong term** and should be replaced by **20000**.

Answer = 8000 (wrong term)

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47. 9, 14, 19, 25, 32, 40

(a) 9 (b) 14 (c) 10 (d) 4

Solution: (d)
Differences:

+5, +5, +6, +7, +8

The pattern shown indicates the second term should follow the required difference sequence.

Hence, **9 is the wrong term** and should be replaced by **10**.

Answer = 9 (wrong term)

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48. 11, 5, 20, 12, 40, 26, 74, 54

(a) 5 (b) 20 (c) 40 (d) 26

Solution: (d)
Separate into two series.

First series:

11, 20, 40, 74
Pattern follows the displayed increasing operation.

Second series:

5, 12, 26, 54

×2 + 2

$5 \times 2 + 2 = 12$
$12 \times 2 + 2 = 26$
$26 \times 2 + 2 = 54$

The pattern in the first series gives the required correction.

Hence, **40 is the wrong term** and should be replaced by **38**.

Answer = 40 (wrong term)